PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Postoperative care in ICU versus non-ICU after head and neck free flap surgery: A systematic review and meta-analysis
AUTHORS	Mashrah, Mubarak; Aldhohrah, Taghrid; Abdelrehem, Ahmed; Sabri, Bahia; Ahmed, Hyat; Al-Rawi, Natheer; Yu, Tian; Zhao, Shiyong; Wang, Liping; Ge, Linhu

VERSION 1 – REVIEW

REVIEWER	Joseph, Sheela Flinders University, NTRCS
REVIEW RETURNED	30-Jul-2021

GENERAL COMMENTS	The review topic is pertinent in informing guidelines for cost effective post operative care in head and neck flap surgery management without compromising on high quality nursing care. Well done! Points for correction: The format of the article will need to be altered to the updated version of PRISMA 2020 checklist There are some typing and grammatical errors for e.g. on page 8 "The titles and abstracts of the remaining 1754 articles were screened and 1734 records were excluded due to being topic off or non-English studies" - the intended meaning probably was that 1734 records were off topic not topic off. Please review the entire article carefully and make appropriate corrections on grammar and expression. Table 1 - column width needs to be adjusted, Abbreviations used must be clearly explained and included with the title of the table. The results described in the table should be described in words as well. Please check and amend the formatting, column titles and title description of all the tables The meta analysis and figures need to be explained in detail in the results section and described adequately in the title of the figures. References should be included appropriately in the description and in the tables. The limitations of the systematic review could be described more clearly in the discussion section Best wishes

REVIEWER	Poder, Thomas University of Montreal, School of Public Health
REVIEW RETURNED	28-Aug-2021

GENERAL COMMENTS	Thanks for this interesting study.

	As I was asked to review on statistical methods and analyses, I will merely focus on this. This being said, statistical methods were appropriate. I2 statistics is not provided in graph as well as which model is used; please add this information. Other comments: What could be the savings? Maybe could refer to some studies (e.g., https://pubmed.ncbi.nlm.nih.gov/23182889/) Why head and neck are not in the PICO? No T and S in the PICOTS… A limit is that few databases were searched, and no gray literature. Should be discussed. Why did you exclude Chinese studies? For instance, some of the co-authors are Chinese and may have read these studies… Would be great to reformat tables (hard to read).

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Sheela Joseph, Flinders University
Comments to the Author:
The review topic is pertinent in informing guidelines for cost effective post operative care in head and neck flap surgery management without compromising on high quality nursing care. Well done!
Our response:
Thank you very much for your positive feedback and for your kind help. All the items you mentioned above are now addressed.

Points for correction:
The format of the article will need to be altered to the updated version of PRISMA 2020 checklist
Our response:
Thank you very much for your time and effort spent on reviewing our manuscript and for your kind help. The format of the article is now adhered to PRISMA 2020 checklist. We changed the whole article format to match PRISMA 2020 checklist as suggested.

There are some typing and grammatical errors for e.g. on page 8 "The titles and abstracts of the remaining 1754 articles were screened and 1734 records were excluded due to being topic off or non-English studies" - the intended meaning probably was that 1734 records were off topic not topic off. Please review the entire article carefully and make appropriate corrections on grammar and expression.
Our response:
Thank you very much for your time and effort spent on reviewing our manuscript and for your kind help. All grammar errors are now corrected. We checked the whole text carefully for language edition and grammar correction.

Table 1 - column width needs to be adjusted, Abbreviations used must be clearly explained and included with the title of the table. The results described in the table should be described in words as well.
Our response:

Thank you very much for your important suggestions and comments. Table 1 - columns width is now adjusted; Abbreviations are now clearly explained at the end of the Table. The results described in the table are also described in words under each outcome.

Please check and amend the formatting, column titles and title description of all the tables.
Our response:
Thank you very much for your comment. Table formatting, column titles and title description of all the tables are addressed.

The meta-analysis and figures need to be explained in detail in the results section and described adequately in the title of the figures.
Our response:
Thank you very much for your suggestion. The meta-analysis and figures are now explained in detail in the results section and described adequately in the title of the figures.

References should be included appropriately in the description and in the tables.
Our response:
Thank you very much for your comment. References are now included in the description and in the tables.

The limitations of the systematic review could be described more clearly in the discussion section
Our response:
Thank you very much for your time and effort spent on reviewing our manuscript and for your kind help. The limitations of the current study are now clearly described as follow:
((Some limitations of this study should be declared. First, only one RCT was included; however, conducting a well-designed RCT in the field of reconstructive surgery appear to be difficult because there are many methodological and practical challenges of the surgical trials, such as recruitment, patient and surgeon preferences, feasibility of blinding, surgical quality control, and standardization of interventions. Second, several cofounders that might have an effect on the postoperative outcomes were not evaluated in the included studies such as the level and specialist training of the staff, nurse to patient's ratio, amount of intraoperative fluid used, use of vasopressors and anticoagulant … etc. For an optimum comparison of different nursing protocols with the least bias, homogeneous sample of patients who were operative for same flap type and flap size, same recipient site, same perioperative care. Third, although there was a consensus among the included studies in that ICU care increases the total hospital cost of care, no meta-analysis was performed due to both missing and heterogeneity in the reported data. Fourth, the non-English studies were excluded from the present study which may result in what is called language bias; however, several studies shown that exclusion of non-English publications from systematic reviews did not significantly affect the results of the meta-analyses [32,33]. In addition, studies published in non-English languages are difficult to locate, and may require translation, which will increase costs and delay the conclusion of a review.))

Best wishes

Reviewer: 2
Dr. Thomas Poder, University of Montreal
Comments to the Author:
Thanks for this interesting study.
Our response:
Thank you very much for your positive feedback and for your kind help.

As I was asked to review on statistical methods and analyses, I will merely focus on this.

This being said, statistical methods were appropriate. I2 statistics is not provided in graph as well as which model is used; please add this information.
Our response:
Thank you very much for your time and effort spent on reviewing our manuscript and for your kind help. The I2 and the used model for each outcome of interest has been added in the Table 3 and in the result section in our previous submission. Based on you suggestion we also added the I2 and model in the Figures descriptions.

Other comments:
What could be the savings? Maybe could refer to some studies (e.g., https://pubmed.ncbi.nlm.nih.gov/23182889/).
Our response:
Thank you very much for your comment. We agree with the reviewer regarding that the implantable Doppler system is more effective than the conventional methods used to monitor free flap perfusion; particularly in poorly accessible flaps. The mean flap salvage rate with the implantable Doppler was 21 percentage points higher (81.4 vs. 60.4). The above mentioned reference and the following sentence are now added in the discussion section:
However, to improve the success rate of free flaps; particularly, in cases with poorly accessible flaps, an implantable Doppler would provide a higher flap salvage rate compared to the conventional monitoring method (81.4 % vs. 60.4%)).[27]. (Please see page ??? in the discussion part)

Why head and neck are not in the PICO?
No T and S in the PICOTS…
Our response:
Thank you very much for your comment. T and S are now added to the revised form, please see material part page 5.

A limit is that few databases were searched, and no gray literature. Should be discussed.
Why did you exclude Chinese studies? For instance, some of the co-authors are Chinese and may have read these studies…
Our response:
Thank you very much for your comment. Several databases can be searched for grey literature, the common Grey databases are Open Grey, WONDER and Scopus. In the current study, we used Scopus database to search for gray literature. Scopus is the largest abstract and citation database of peer-reviewed literature: scientific journals, books, and conference proceedings. It includes over 7.2 million conference papers from over 88,800 worldwide events, peer-reviewed papers, and proceedings and patents from the United States, United Kingdom, Europe, Japan, and the World Intellectual Property Organization. (Please see this link https://onlinelibrary.wiley.com/doi/10.1111/jebm.12266).
In addition, we have searched in (clinicaltrials.gov; www. centerwatch.com/clinical trials; www.clinicalconnection.com).
Regarding the second part of your question why Chinese studies were not included?
Review authors may want to search without language restrictions and decisions about including reports from languages other than English may need to be taken on a case-by-case basis. We agree with the reviewer in that the inclusion of studies published in languages other than English could avoid bias (language bias), however, if Chinese studies were included then we should include studies written in languages other than English and Chinese.
We added this point as one of the limitations of the current study as follow:
Fourth, the non-English studies were excluded from the present study which may result in what is called language bias; however, several studies showed that exclusion of non-English publications from systematic reviews did not significantly affect the results of the meta-analyses [32,33]. In

addition, studies published in non-English languages are difficult to locate, and may require translation, which will increase costs and delay the conclusion of a review.

Would be great to reformat tables (hard to read).
Our response:
Thank you very much for comment. Tables are now checked and carefully reformatted

Reviewer: 1
Competing interests of Reviewer: None

Reviewer: 2
Competing interests of Reviewer: None

VERSION 2 – REVIEW

REVIEWER	Poder, Thomas University of Montreal, School of Public Health
REVIEW RETURNED	15-Nov-2021

GENERAL COMMENTS	Thanks for this interesting study. In the bullets: “was the high heterogeneity between” Scopus is not really grey literature… S in PICOTS: trial and not trail; T in PICOTS: hours. Generally, there is many typo errors. Please check.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 2

Dr. Thomas Poder, University of Montreal

Comments to the Author:

Thanks for this interesting study.

Our response:

Thank you very much for your positive feedback.

In the bullets: “was the high heterogeneity between”

Our response:

Thank you very much for your valuable comment. The statement “was the high heterogeneity between” in the Strengths and limitations is now added.

Scopus is not really grey literature…

Our response:

Thank you very much for your valuable comment. Totally agree with you in that “Scopus is not really grey literature”.

S in PICOTS: trial and not trail; T in PICOTS: hours.

Our response:

Thank you very much for your valuable comment. All typing errors are now corrected as suggested.

Generally, there is many typo errors. Please check.

Our response:

Thank you very much for your valuable comment. All typo errors are now checked and corrected as suggested.